

GAURAB ACHARYA

Vancouver, BC, Canada

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EDUCATION

University of British Columbia

Bachelor of Applied Science in Biomedical Engineering — Specialized in Systems and Signals

Sep. 2019 – Apr. 2025

Vancouver, British Columbia

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL, Java, C++, C, HTML, CSS, R, Swift

Frameworks: React, Next.js, Jest, Spring Boot, Flask, TensorFlow, JUnit, R Shiny, Tailwind CSS

Technologies: Docker, Apache Airflow, Jenkins, PostgreSQL, Oracle, Clickhouse, Figma Grafana, Kibana

EXPERIENCE

QuoteMedia

May 2024 – Aug. 2024, May 2022 – Apr. 2023

Software Developer Intern

Vancouver, BC

- Built scalable API endpoints using **Java**, **SQL**, and **Spring Boot** to support market data delivery to major financial clients
- Optimized query performance 10x by migrating legacy databases (**Oracle to PostgreSQL**) and tuning schema
- Contributed to a cross-functional pipeline migration project using **SQL**, **Python**, and **Apache Airflow**, increasing data aggregation speed by **20x** for analytics workflows
- Modernized frontend chart endpoints to enhance responsiveness and load time, improving merchant-facing user experiences
- Designed and implemented a performance testing pipeline using **Vegeta** and **Apache Airflow**, visualized with **InfluxDB** and **Grafana** for rapid iteration and feedback loops
- Built internal dashboards with **Kibana** and **Grafana** to summarize API speed, and usage, aiding rapid decision-making

ProCogia

May 2023 – Sep. 2023

R Developer Intern

Vancouver, BC

- Developed an **R** package integrating **OpenAI ChatGPT API** to analyze and debug R scripts in <5 seconds, increasing efficiency in developing data driven projects
- Engineered system prompts and conversational memory to improve AI response consistency by 90+% across diverse use cases
- Took full ownership of package architecture, testing, and deployment

PROJECTS

AI-Powered Medical Chatbot  | *React, Next.js, Tailwind CSS, Python, LangChain, Flask, Pinecone, Google Gemini*

- Built a modern, responsive web UI using **Next.js** and **Tailwind CSS**, enabling users to receive verifiable, source-linked responses in real-time while reducing hallucinated outputs through **Retrieval Augmented Generation (RAG)**
- Designed **Figma wireframes**  and translated them into dynamic UI components with component-driven architecture
- Improved citation quality by **20%** and reduced hallucinated responses across **1,000+ queries** using **Gemini-2.0** and RAG
- Indexed the Gale Encyclopedia of Medicine using the vector database, **Pinecone**, and **Hugging Face embeddings**, then deployed it as a **Flask API**  to deliver real-time, context-aware responses for seamless integration with a React-based chatbot interface

UBC Capstone Medical Augmented Reality Device | *Unity, C#, ARKit*

- Developed an iOS app using **Unity** and **OpenXR** to render vein images for a Medical Device Startup, Vantech Medical, using **Apple ARKit's** image tracking and projection
- Integrated a **Google cloud hosted** infrared (IR) vein detection model, achieving real-time rendering with less than 1-second latency
- Conducted **UI testing** with healthcare workers, gathering user feedback to improve functionality and user experience

Sneaker Image Recognition App  | *Hack the North Winner*  | *Python, TensorFlow Lite, Swift*

- Trained an image classification machine learning model using **Python** and **TensorFlow Lite**, to recognize Jordan sneaker models
- Built an iOS application, using **Swift**, allowing users to upload photos to recognize the model of Jordan sneakers
- Contained the machine learning model on device using the **HOT-G API**, to ensure data security and speed

LEADERSHIP / EXTRACURRICULAR

Basketball / Coaching

Apr. 2016 – Present

RBL Basketball , *Drive Basketball* , *Strathcona Community Centre*, *Dreamers Basketball* 

- 4x British Columbia Provincial Basketball Championship All-Star
- Coached over 1500 players in basketball camps and teams, working with four different clubs in Vancouver for over 7 years